

Claire Mundi

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University of Wisconsin-Madison
Department of Atmospheric & Oceanic Sciences

Education

- 2022–2026 **Ph.D.** Atmospheric & Oceanic Sciences, University of Wisconsin, Madison, WI
Dissertation Title: "Quantifying Seasonal, Hemispheric, and Interannual Variations in Cyclone Impacts on Sea Ice"
- 2020–2022 **M.S.** Atmospheric & Oceanic Sciences, University of Wisconsin, Madison, WI
- 2016–2020 **B.A.** Physics, Geosciences, Skidmore College, Saratoga Springs, NY
Summa Cum Laude | *Phi Beta Kappa*, *Sigma Pi Sigma*, *Pi Mu Epsilon* Honor Societies

Research Experience

- 2020– **Graduate Research Assistant**
[The Atmospheric Radiation and Climate Research Group](#)
University of Wisconsin-Madison (Advisor: Dr. Tristan L'Ecuyer)
Analyzed the role of intense cyclones on the marginal ice zone in both hemispheres using a combination of satellite observations, reanalyses, and climate model output.
- 2019–2020 **Undergraduate Research Assistant**
Skidmore College (Advisor: Dr. Greg Gerbi)
Investigated particle transport pathways for two species of sea snail larvae in the Delaware Bay Region using a coupled hydrodynamic model (ROMS) and a Lagrangian particle-tracking model (LTRANS).
- 2018 **Scripps Undergraduate Research Fellowship (SURF)**
Scripps Institution of Oceanography (Advisor: Dr. Dan Lubin)
Determined 3 years of cloud phase properties over the Greenland Ice Sheet using ground-based micropulse lidar signals from Summit Station.

Awards & Honors

- 2020 Eleanor A. Samworth Prize for Excellence in Physics (Skidmore College)
- 2016–2020 Porter Presidential Scholar in Science and Mathematics (Skidmore College)

Publications

 [Google Scholar](#)  [0009-0007-4477-0698](#)

Journal Articles

- J1. **Mundi, Claire** & L'Ecuyer, T. Is the Modern Arctic Marginal Ice Zone More Susceptible to Summer Cyclones? *Journal of Climate* **38**, 403–422 (2025).
- J2. **Mundi, Claire**, L'Ecuyer, T. & DuVivier, A. Have Impacts of Intense Arctic Cyclones on Summer Sea Ice Reached a Maximum? *Geophysical Research Letters* **52**, e2025GL117848 (2025).

Working papers

- W1. **Mundi, Claire**. & L'Ecuyer, T. Intense Storms Increase Global Ice Area. *Nature Communications*, Under Review. 2026.

Presentations

Talks

- T1. **Mundi, Claire** & L'Ecuyer, T. *An Observed Seasonal Symmetry in Intense Cyclone Impacts on the Global Marginal Ice Zone* AGU Annual Meeting (New Orleans, LA, USA). Dec. 2025.
- T2. **Mundi, Claire** & L'Ecuyer, T. *Observed and Predicted Changes in Arctic Sea Ice Response to Intense Summer Cyclones* AMS 18th Conference on Polar Meteorology and Oceanography (Denver, CO, USA). May 2025.
- T3. **Mundi, Claire** & L'Ecuyer, T. *How Do Intense Summer Cyclones Impact the Arctic Marginal Ice Zone in the CESM2 Large Ensemble?* NCAR Polar Climate Working Group Meeting (Denver, CO, USA). Mar. 2025.

Posters

- P1. **Mundi, Claire** & L'Ecuyer, T. *An Observed Transition to More Marginal Sea Ice Loss Due to Arctic Summer Cyclones* AGU Annual Meeting (Washington, D.C., USA). Dec. 2024.
- P2. **Mundi, Claire** & L'Ecuyer, T. *Role of Cyclone-Induced Near-Surface Winds and Sea Surface Temperature Changes on Arctic Sea Ice* AMS 17th Conference on Polar Meteorology and Oceanography (Madison, WI, USA). Aug. 2022.
- P3. Beckley, I., **Mundi, Claire**, L'Ecuyer, T. & Martin, J. *Analysis of the Kinematic and Radiative Impacts on Sea Ice of a Robust Late-Summer Cyclogenesis Event in the Arctic*. AMS Annual Meeting, Conference on Mesoscale Processes (Remote). Jan. 2022.
- P4. **Mundi, Claire**, Wilson, J., Gerbi, G., Fuchs, H., Hunter, E., Chant, R. & Wilkins, J. *Particle pathways in the mid-Atlantic bight*. AGU Ocean Sciences Meeting (San Diego, CA, USA). Feb. 2020.
- P5. **Mundi, Claire** & Lubin, D. *Frequency of Radiatively Active Liquid Water Clouds Above the Greenland Ice Sheet*. Summer Symposium, Scripps Institution of Oceanography (La Jolla, CA, USA). Aug. 2018.

Outreach

- O1. **Mundi, Claire.** *Observed and Predicted Changes in Arctic Sea Ice Response to Intense Summer Cyclones: Research Overview* FIRE Climate Computing, University of Maryland (Remote). June 2025.
- O2. **Mundi, Claire.** *Physics Alumni Research Talk.* Skidmore College (Remote). Feb. 2023.

Mentorship and Teaching

- 2024–2026 **IARPC Collaborations Mentorship Program**
- 2020–2025 **AOS Mentorship Program**, University of Wisconsin, Madison, WI
Mentored undergraduate students through monthly meetings.
Led department-wide networking and writing workshops.
- 2017–2020 **Peer Academic Coach**, Skidmore College, Saratoga Springs, NY
General Physics I, General Physics II, Modern Physics, Thermal and Statistical Physics, Calculus II, Atmosphere and Ocean Dynamics, Mathematical and Computational Methods in Physics
- 2019–2020 **Peer Mentor for First-Year Seminar**, Skidmore College, Saratoga Springs, NY
The Deep Seafloor: Facilitated discussions on ocean observations and exploration.

Academic Service and Outreach Experience

Journal Reviewer

- 2026 Nature Communications
- 2026 The Cryosphere
- 2025 Journal of Geophysical Research - Atmospheres

Volunteer Experience

- 2020–2025 **AOS Graduate Student Association**, University of Wisconsin, Madison, WI
Performed public science demonstrations to over 1,000 students across several events including the WI Science Festival and UW Science Expeditions.
Alumni Engagement Board Representative (2021–2022)
- 2022–2024 **STORM REU**, University of Wisconsin, Madison, WI
Organized professional development events for three cohorts of visiting undergraduate summer research students.
- 2022–2026 *Letters to a Pre-Scientist* Volunteer

Served as a STEM pen-pal to four middle school students from low-income areas, addressing topics such as education pathways, overcoming barriers, and various career opportunities in STEM.

2023–2026 *Adopt-a-Physicist* Volunteer
Responded to over 50 questions per year from high school physics students about Arctic climate, graduate programs, and careers in science.

Trainings and Workshops

2025	Scalable and Computationally Reproducible Approaches to Arctic Research , National Center for Ecological Analysis and Synthesis, Santa Barbara, CA
2024	NOAA Women in Sciences Leadership Workshop , Geophysical Fluid Dynamics Laboratory, Princeton, NJ
2023	CESM Tutorial , National Center for Atmospheric Research, Boulder, CO

Last updated: April 2026